INCH-POUND

MIL-PRF-55339/52A
w/AMENDMENT1
24 January 2025
SUPERSEDING
MIL-PRF-55339/52A
30 April 2015

PERFORMANCE SPECIFICATION

ADAPTER, CONNECTOR, ELECTRICAL, COAXIAL, RADIO FREQUENCY,
(BETWEEN SERIES SMA TO N)

This specification is approved for use by all
Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall
consist of this specification sheet and MIL-PRF-55339.

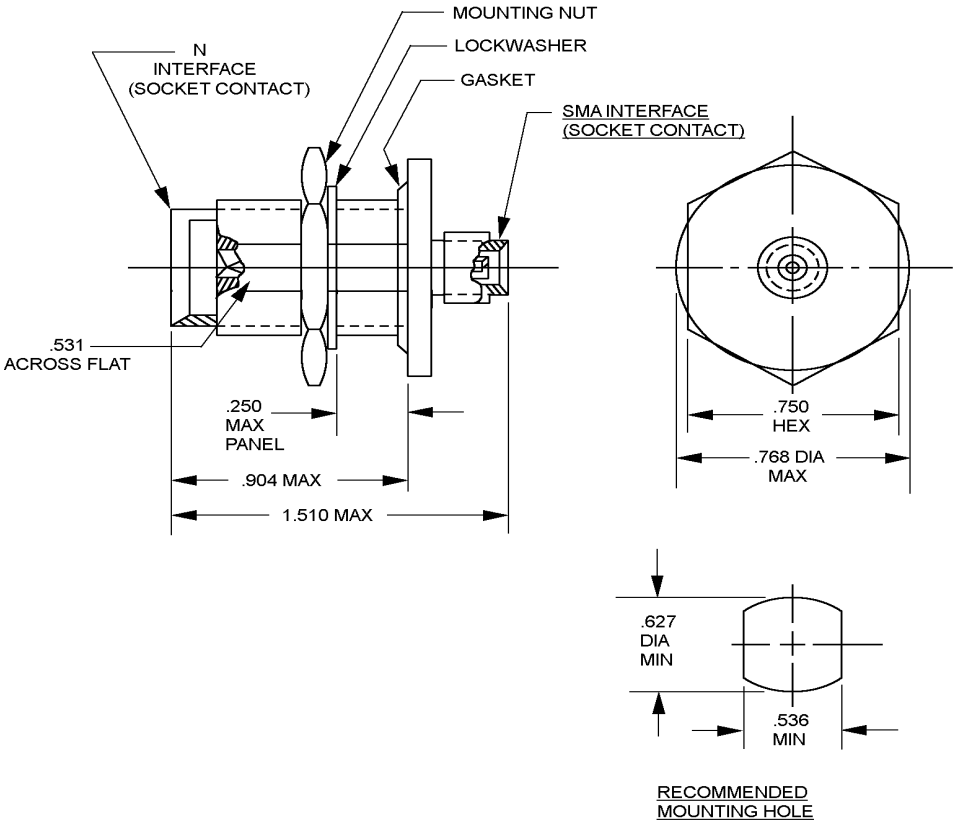
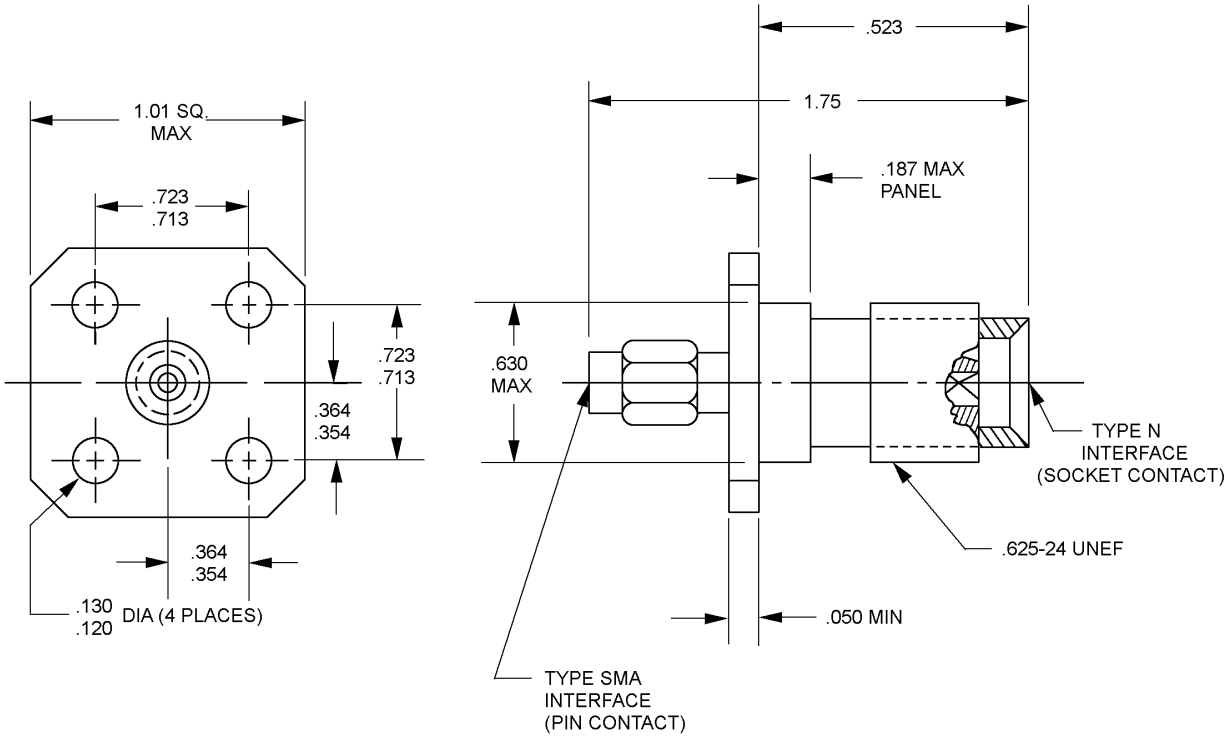


FIGURE 1. General configuration.

AMSC N/A
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FSC 5935

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NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified, tolerance is ± .015 (0.38 mm).
3. All undimensioned pictorial configurations are for reference purposes only.
4. N and SMA interfaces shall be in accordance with MIL-STD-348.

FIGURE 1. General configurations – Continued.

ENGINEERING DATA:

Nominal impedance: 50 ohms.

Frequency range: 18 GHz.

Voltage rating:

335 V rms at sea level.

85 V rms at 70,000 feet.

Operating temperature range: -65°C to +165°C.

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REQUIREMENTS:

Dimensions and configuration: See figure 1 and MIL-STD-348.

Center contact retention:	Series SMA	Series TNC
Axial force (pounds, minimum)	6	6
Torque	Not applicable	Not applicable
Coupling proof torque	15 inch-pounds, minimum	
Force to engage and disengage:	Series SMA	Series TNC
Longitudinal force (pounds, minimum)	Not applicable	3
Torque (inch-pounds, maximum)	2	6
Mating characteristics in accordance With MIL-STD-348:	Series SMA	Series TNC
Center contact (socket):		
Oversize test pin diameter (inch)	.0375 + .0001	.074 minimum
Insertion depth (inch, minimum)	.030/.045	.125
Number of insertions	3	1
Pin finish (microinches)	16	16
Maximum test pin (insertion force test):		
Steel test pin diameter (inch, minimum)	.0370 + .0001	.066
Pin finish (microinches)	16	16
Insertion force (pounds, maximum)	3	2
Insertion depth (inch, minimum)	.050/.075	.125
Minimum test pin (withdrawal force):	Series SMA	Series TNC
Steel test pin diameter (inch)	.0355 - .0001	.063
Pin finish (microinches)	16	16
Withdrawal force (ounces, minimum)	1	1
Insertion depth (inch, minimum)	.050/.075	125

Permeability: Less than 2.0, air = 1.0.

Seal:

Pressurized: Not applicable.

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Weatherproof: Not applicable.

Insulation resistance: 5,000 megohms, minimum.

VSWR:

1.06 + .005 F (GHz); dc to 12.4 GHz.

.83 + .023° F (GHz); 12.4 to 18 GHz.

RF leakage (total): -65 dB, minimum, 2 to 3 GHz.

RF insertion loss: .18 dB, maximum, at 9 GHz.

Dielectric withstanding voltage: 1,000 V rms at sea level, minimum.

Contact resistance (milliohms, maximum):

Contact	Initial	After environmental
Center	4.1	6.0
Outer	2.2	Not applicable

Vibration, high frequency: Interruptions: 1 µs, maximum, MIL-STD-202-204, test condition D.

Shock: MIL-STD-202-213, test condition I.

Thermal shock: MIL-STD-202-107, test condition C.

Moisture resistance: MIL-STD-202-106.

Insulation resistance: 200 megohms, minimum, within 5 minutes after removal from humidity.

Corona level:

Voltage: 250 V, minimum at 70,000 feet, minimum.

RF high potential withstanding voltage:

RF voltage: 1,000 V rms, minimum, at 5 MHz, minimum.

Salt spray: MIL-STD-202-101, test condition B.

Connector durability: 500 cycles minimum at 12 cycles per minute maximum. The connector shall meet mating characteristics and force to engage and disengage.

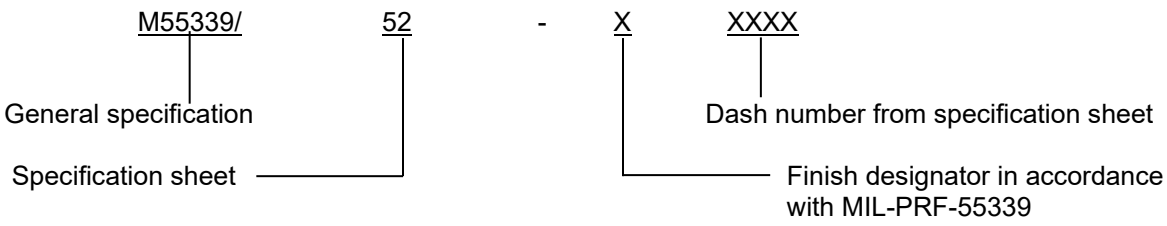
Marking: As specified in MIL-PRF-55339.

Part or Identifying Number (PIN):

M55339/52-30001
M55339/52-50001

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PIN configuration:



Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents: In addition to MIL-PRF-55339, this document references the following:

MIL-STD-202-101
MIL-STD-202-106
MIL-STD-202-107
MIL-STD-202-204
MIL-STD-202-213
MIL-STD-348

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA – CC

Preparing activity:
DLA - CC

(Project 5935-2025-009)

Review activities:
Army - AR, AT, MI
Navy - AS, MC, OS, SH
Air Force – 19

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of information above using the ASSIST Online database at https://assist.dla.mil.

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